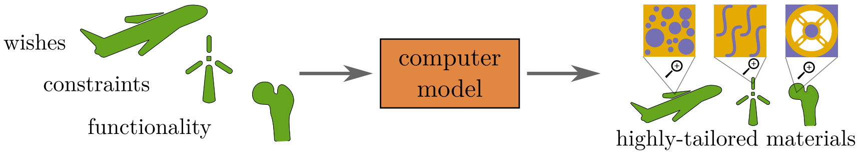
Materials and computer models – avenues for progress

The answer: combining a handful of experiments with extensive virtual testing

Design optimization

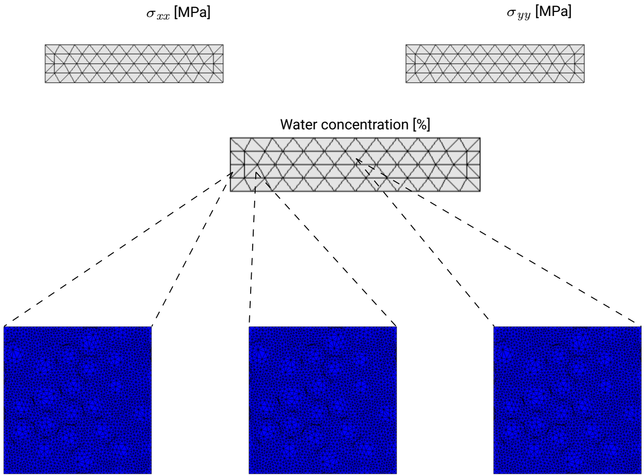
- Design variables can be tweaked at different scales (e.g. microstructures, micro properties)
- Finding the optimum among millions of possible combinations $\Rightarrow$ more efficient designs



Example — Aging of fiber-reinforced polymers

Now for our final trick, we combine these two scales into a single model:

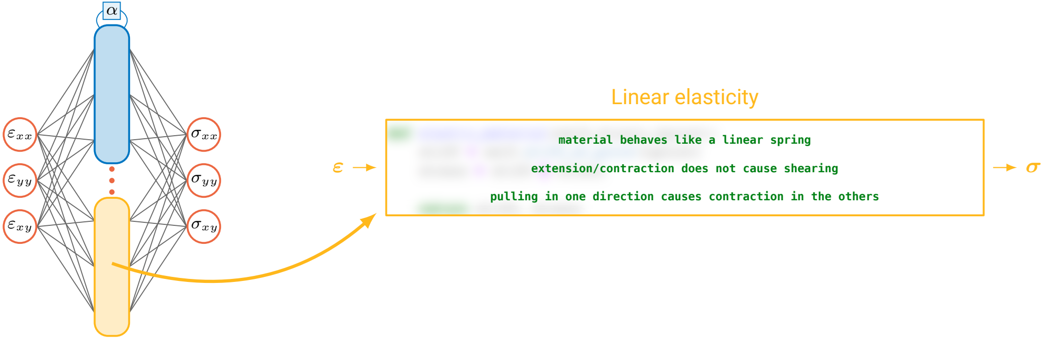
- FEM at both scales, so a lot of FEM models to run...



Physically Recurrent Neural Networks

Our solution is to go for a hybrid machine learning model:

- Substitute some of the artificial neurons for cells with real physical models
- Leave the rest of the model intact and train it in the same way as normal ML models
- How are the models added? By using material model code we had been using for decades



[Maia *et al.*(2023), *CMAME*]

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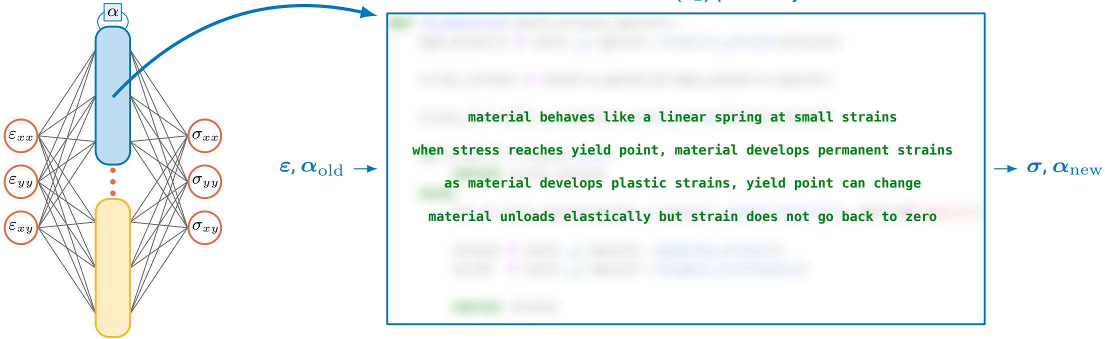
Von Mises (J_2) plasticity



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Going from prior to posterior

It is often impossible to compute the posterior exactly, so we approximate it: